Department of Software Engineering
Government College University, Faisalabad.

Assignment 1, February-2026, session 2025-29, **Submission Date: Last week:February 2026** (in Lab class)

Class	: BS (SE)	**Semester**	: 2nd
Course Title	: Object Oriented Programming	**Course Code**	: **CCC** – 302
Instructor	: Dr. M. Awais	**Credit Hour**	: 04 (3 – 1)

Question # 1: Write down a programs as per following specifications. The class definition should be given in header file, named "bank.h". The member functions detail should be given in separate "bank.cpp" file. There should be four different files named client1.cpp, client2.cpp client3.cpp and manager.cpp. First the Manager should create three clients and each client should at least should first view its account then deposit some amount in the account and then view the account then withdraw amount more than amount that exists in the account. The messages given to the client should be invalid operation. Then the client should withdraw valid (less than amount in the account) amount from the account and lastly again view the account. The Manager should once view all the data of clients.

Moreover, the Manager object creation must be protected by password to avoid easy creation of Manger object.

Bank Management System having the following Member variables and functions

1. Client Data (Member variables)
   i. Client Name (used as username)
   ii. Password (characters should not be shown or as *)
   iii. Client CNIC (must be unique)
   iv. Client Bank Account number (must be unique for the clients)
   v. Client Cash in account
   vi. Manager username (Must be only one for Clients)
   vii. Manager password (Must be only one for Clients)
2. Operations that can be performed by the Client on the data
   i. View his Bank Account
   ii. Can change his Name if client can provide correct CNIC
   iii. Can deposit in the account
      a. It will update the Cash in the clients account
   iv. Can withdraw from his account by verifying the username and password
      a. It is only possible if the account has the requested amount of money
      b. It will update the Cash in the account by subtracting the withdrawn amount
   v. The new account can be only created by one Manager account, which is guarded by username and password
   vi. The Manager can check the data of all the clients

SUBMISSION GUIDELINES

THE STUDENTS WILL SUBMIT ASSIGNMENT IN HARD AND SOFT FORM (muhammadawais@gcuf.edu.pk), SUBJECT OF EMAIL SHOULD BE ONLY "**OOP/ EVE OR MOR/SECTION A OR B, ROLL#**"